Machine Perception Report: PerceiveAndRejoice

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ABSTRACT

In this study, we assessed the performance of various human motion prediction models on a subset of the AMASS dataset. The best-performing model was a Graph Convolutional Network (GCN) enhanced with an attention mechanism. It achieved a value of 1.69 for the joint angle discrepancy metric on the public part of the test set. This model demonstrated superior accuracy and on-par training efficiency compared to other considered models.

1 INTRODUCTION

Human motion prediction is the task of forecasting future 3D human body poses based on a sequence of prior poses. In this study, we used a subset of the AMASS database [4]. The training data was divided into sequences of 144 frames, with the first 120 frames serving as the input sequence and the remaining 24 frames used for prediction evaluation. Each frame represents a human body pose with 15 joints, each described by a 3x3 rotation matrix. For models that work with 3D joint coordinates, we applied forward kinematics to convert the AMASS data. Our model consists of a Graph Convolution Network (GCN) whose input was the result of a motion attention mechanism, allowing us to focus on relevant past frames, as well as the encoding of the last motion.

2 CONSIDERED MODELS

Due to the temporal nature of human motion prediction, many previous approaches use Recurrent Neural Networks (RNNs). However, as noted by [7], RNNs suffer from discontinuities in predicted frames and error accumulation. They attempted to model velocities instead of poses to mitigate this, but discontinuities remained between the last observed and the first predicted frame.

The introduction of the transformer architecture has provided an effective solution for motion prediction. Its attention mechanism allows the model to dynamically focus on different parts of the input sequence and learn long-range dependencies. Attention-based models have shown improvements in both short and long-term motion prediction by capturing similarities between current and historical motion contexts [1, 2]. Despite their effectiveness, these models still require structural priors and are complex and difficult to train.

Therefore, efforts have been made to reduce model complexity, demonstrating that high predictive power is not strictly dependent on high model complexity. Guo et al. [3] present SiMPLe, a model based on a stack of multi-layer perceptron (MLP) blocks with residual connections that operate in the frequency domain. This network achieves state-of-the-art results with a fraction of the parameters needed for previous models.

Another main approach to modeling human motion is the use of Graph Neural Networks. They take advantage of the graph structure of the skeleton, which can be viewed as a hierarchical-like tree where the joints are vertices. Most research papers employ some sort of graph convolutions (whether they are spatial, temporal, spatial-temporal, trajectory, etc). Sofianos et al. [8] propose the Space-Time-Separable Graph Convolutional Network (STS-GCN), which integrates temporal evolution and spatial joint interactions within a unified framework. Despite having a better performance than the model we opted for when taking 10 frames as input, it didn't handle the longer input sequences we had (120 frames) as well.

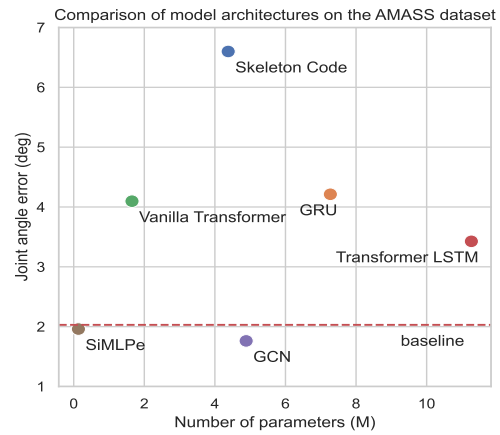

Figure 1: Model architecture comparison.

3 METHOD

One of the seminal works on the use of GCN models for human motion prediction was conducted by [6]. To model the temporal dependencies, they propose representing human motion in trajectory space instead of pose space. This can be achieved by using the Discrete Cosine Transform (DCT) to encode temporal information.

The method we used in the following model was based on the paper History Repeats Itself by Mao et al. [5]. Building on their prior work [6], [5] add an attention mechanism to their GCN model. The main idea behind the method is that humans tend to repeat their actions. Hence, we aim to find relevant historical motion sequences to help us predict future ones. We cut the past observations into subsequences and employ motion attention to exploit the similarities between them and the current observation. This approach leverages the repetitive nature of human motion, allowing past observations to serve as a powerful prior for predicting future movements.

We will now define the model more formally. Let $\mathbf{X}_{1:N} = [\,\mathbf{x}_1, ..., \mathbf{x}_N]$ be the motion history with N:= 120 consecutive human poses each having dimension $\mathbf{x}_i \in \mathbb{R}^K$ where K := 135 = 15*3*3 are the angles associated to each of the 15 joints. Our goal is to predict the future sequence $\mathbf{X}_{N+1:N+T}$ where T := 24. We define a kernel size of M and divide our motion history into M-T+1 sub-sequences of size M+T. This can be interpreted as using the past M frames to predict the future T. In our case, we set M := 42. We will then use two 1D convolutional networks to map on the one hand our last

subsequence of length M to a query $q \in \mathbb{R}^d$, and the first N-M-T+1 subsequences of length M to our keys vectors $k_i \in \mathbb{R}^d$ where d is the dimension of our hidden features which we set to 256. We can then compute the attention score:

$$a_i = \frac{qk_i^T}{\sum_{i=1}^{N-M-T+1} qk_i^T} \tag{1}$$

Our value matrices $V_i \in \mathbb{R}^{135\times(M+24)}$ are the DCT coefficients of a joint. The output of our attention model will then be the values weighted by the attention. This provides us with an initial estimate of the future motion.

The input of the first layer of our graph convolutional network is the concatenation of the attention model's output and the DCT coefficients. It is a matrix of dimension: Nr. of pose parameters (135) x 2*(Size of Kernel(42)+Length of Sequence to be Predicted (24)). Our GCN architecture contains at each layer p: a trainable adjacency matrix A^p, a trainable weight matrix W^p. The output of the layer p can be computed with the following equation:

$$H^{p+1} = \sigma(A^p H^p W^p) \tag{2}$$

Our architecture contained a total of 12 residual blocks, each having 2 convolutional layers. Finally, we use the Inverse Discrete Cosine Transformation to reconstruct our predicted sequence.

4 EVALUATION

We chose to mostly follow the hyperparameters defined in the paper History Repeats Itself by Mao et al. [5]. They seemed to work quite well when we completed our ablation studies. One major difference is that we trained our model for 12000 epochs, since performance continuously improved as we increased our number of epochs.
We performed ablation studies on the number of residual blocks in our model. We observed the joint angle loss on the validation set after 50 epochs. The default value of 12 seemed to be ideal. Additionally, we also conducted ablation studies on the number of hidden features where the default value was once again ideal.

NUM STAGES	joint_angle
8	3.070
10	3.049
12	3.012
14	3.032
16	3.117

Table 1: Validation Joint Angle Loss for Different Num Stages after 50 Epochs

In order for our model to be more stable during training, we considered implementing gradient clipping. Initial results indicated that gradient clipping would be helpful. However, as we increased the number of epochs they seemed to harm our model's performance. Finally, we discovered that increasing our kernel size despite having sub-optimal performance early on proves to be optimal as we increase the number of epochs. Thus, we set our kernel size to 42.

In addition, we also tried using the Stochastic Gradient Descent (SGD) optimizer and obtained a joint angle loss of 4.255 after 50 epochs. This is noticeably inferior to the default ADAM optimizer with a joint angle loss of 3.012 after 50 epochs.

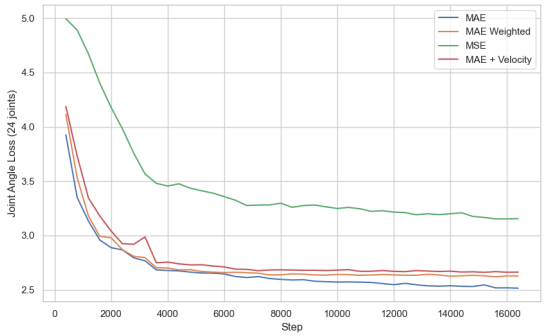

Figure 2: Ablation on loss functions based on the GCN model trained for 500 epochs.

5 DISCUSSION

For GCN, the number of DCT components determines the level of detail in representing motion data. Selecting too few components results in a coarse representation while selecting too few may capture noise in the data. The final value of 42 is relatively high compared to [8] and may lead to excessively optimistic results on the public test set that may noticeably degrade on the private one.

A crucial component of training is the objective. As demonstrated in Fig. 2, Mean Absolute Error (MAE) loss results in the best value of the joint angle discrepancy metric. However, the pattern we observed by calculating the metric on a rolling basis from the first joint to the last, states that different joints contribute non-uniformly to the metric. While joints from 0 to 10 are responsible for $\tilde{2}$0% of its value, joints from 10 to 19 that correspond to the most agile body parts (e.g., foot, head, and shoulders) are responsible for $\tilde{4}$5%, and joints 19-24 corresponding to arms - 35%. This motivates the introduction of another loss - weighted MAE - where errors on the most problematic joints are penalized more than others which experimentally was proven to worsen the result in the considered setup. Motivated by prior work ([3]), we add a velocity term to the MAE. However, the original promise of this loss component is to improve long-term predictions while not severely degrading short-term ones. Predicting the 24-frame horizon does not benefit from it while slightly worsening the result. Finally, MSE loss is shown to be the worst which could be explained by the presence of sudden movements in the dataset which will be treated as outliers by this loss and the corresponding errors highly penalized.

Data augmentations such as front-back flipping and random rotations to joints did not lead to a better public score while worsening results on the validation set potentially due to too substantial discrepancies not present in actual data.

6 CONCLUSION

GCN model with attention demonstrated the best performance on our task compared to GRU, Transformer LSTM, SiMPLe, and STS-GCN models. Incorporating ideas from other works, such as data augmentation techniques and different loss functions, did not yield significant improvements in model performance.

REFERENCES

[1] Emre Aksan, Manuel Kaufmann, Peng Cao, and Otmar Hilliges. 2021. A spatio-temporal transformer for 3d human motion prediction. In *2021 International Conference on 3D Vision (3DV)*. IEEE, 565–574.

[2] Deepak Gopinath and Jungdam Won. 2020. Fairmotion-tools to load, process and visualize motion capture data.

[3] Wen Guo, Yuming Du, Xi Shen, Vincent Lepetit, Xavier Alameda-Pineda, and Francesc Moreno-Noguer. 2023. Back to mlp: A simple baseline for human motion prediction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 4809–4819.

[4] Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael J. Black. 2019. AMASS: Archive of Motion Capture as Surface Shapes. In *International Conference on Computer Vision*. 5442–5451.

[5] Wei Mao, Miaomiao Liu, and Mathieu Salzmann. 2020. *History Repeats Itself: Human Motion Prediction via Motion Attention*. 474–489. https://doi.org/10.1007/978-3-030-58568-6_28

[6] Wei Mao, Miaomiao Liu, Mathieu Salzmann, and Hongdong li. 2019. Learning Trajectory Dependencies for Human Motion Prediction. 9488–9496. https://doi.org/10.1109/ICCV.2019.00958

[7] Julieta Martinez, Michael J Black, and Javier Romero. 2017. On human motion prediction using recurrent neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2891–2900.

[8] Theodoros Sofianos, Alessio Sampieri, Luca Franco, and Fabio Galasso. 2021. Space-time-separable graph convolutional network for pose forecasting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 11209–11218.